Unit Test Report

Automated unit and integration (database) test results – Software Test Plan v1.1, §5 test levels 1 and 2

Project	WealthWise – Wealth Management System (Team-02)
Report	Unit Test Report
Date	02-11-2026
Build under test	main @ a2c6973 (Sprint 2 + final fixes, PR #77) + testing-week tests
Environment	Linux test machine · Node.js 22.22 · PostgreSQL 16.15 · Next.js 14.2.35 · TypeScript 5.9 · c8 10.1.3 (V8 coverage) · Chromium via Playwright

Unit tests
73/73
passed

Database integration tests
54/54
passed

Statement coverage
96.9%
src/lib

Branch coverage
85.0%
src/lib

1. Summary

Suite	What it tests	Files	Tests	Passed	Failed	Duration
Unit (`npm test`)	Pure functions, input validation, security rules, layering rules – no database	16	73	73	0	7.7 s
Database integration (`npm run test:db`)	Data-access modules against PostgreSQL 16; each file creates and deletes its own users	21	54	54	0	12.1 s
Total		37	**127**	**127**	**0**	19.8 s

Type-check (`npm run typecheck`): 0 errors. The same suites run in GitHub Actions on every pull request (Ubuntu, Node.js 22, PostgreSQL 16).

> **Added during the testing week (2 Nov):** `loans-edit.test.ts`, `news-refresh.test.ts`, `upload.test.ts`, `users-assets.test.ts`, `validation.test.ts` – 10 unit and 10 database tests covering input-validation boundaries, upload checks, user and asset updates, loan edits, ownership checks and the news-feed refresh path (with a stubbed feed). They raised statement coverage of `src/lib` from 85.4 % to 96.9 %.

2. Unit test suite

Command: `npx tsx --test "tests/unit/**/*.test.ts"`

Test file	Covers	Tests	Passed	Failed	Time (ms)
`architecture.test.ts`	NFR-06 layering	5	5	0	9
`credit-score.test.ts`	FR-19 credit score	5	5	0	2
`csv-parser.test.ts`	FR-14 CSV parser	7	7	0	5
`emi.test.ts`	FR-07 / NFR-02 EMI maths	4	4	0	3
`file-store.test.ts`	FR-17 file store	3	3	0	9
`format.test.ts`	NFR-05 number format	2	2	0	16
`fy.test.ts`	FR-22 financial year	4	4	0	2
`news.test.ts`	FR-25 news parsing	5	5	0	7
`pdf-statement.test.ts`	FR-23 PDF statement	3	3	0	46
`prepayment.test.ts`	FR-11 prepayment maths	5	5	0	4
`projection.test.ts`	FR-21 projection	4	4	0	2
`receipt-fields.test.ts`	FR-16 receipt fields	4	4	0	5
`recommend.test.ts`	FR-20 recommendations	4	4	0	18
`security.test.ts`	SEC-01 / SR-01–SR-09	8	8	0	1506

Test file	Covers	Tests	Passed	Failed	Time (ms)
`upload.test.ts` **(new)**	SR-07 upload checks	4	4	0	5
`validation.test.ts` **(new)**	SR-06 input validation	6	6	0	88

2. Unit test suite – individual results

File	Test	Result	ms
`architecture.test.ts`	there is something to check	**Pass**	1.0
`architecture.test.ts`	UI files never import the database layer	**Pass**	1.5
`architecture.test.ts`	route handlers contain no inline SQL and do not touch pg directly	**Pass**	1.0
`architecture.test.ts`	every route handler sends errors through apiError()	**Pass**	2.0
`architecture.test.ts`	only the data-access layer imports pg	**Pass**	3.5
`credit-score.test.ts`	weights add up to 1	**Pass**	1.1
`credit-score.test.ts`	score is always within 300–900	**Pass**	0.6
`credit-score.test.ts`	a good record scores Excellent	**Pass**	0.1
`credit-score.test.ts`	an overdue installment lowers the score (AC-19)	**Pass**	0.1
`credit-score.test.ts`	more debt relative to assets lowers the score	**Pass**	0.1
`csv-parser.test.ts`	reads quoted fields with commas inside	**Pass**	1.9
`csv-parser.test.ts`	parses day-first dates and rejects impossible ones	**Pass**	0.7
`csv-parser.test.ts`	parses Indian digit grouping	**Pass**	0.3
`csv-parser.test.ts`	HDFC layout: skips preamble, finds columns by name, rejects bad rows with reasons	**Pass**	1.1
`csv-parser.test.ts`	ICICI layout with Debit/Credit columns and ISO dates	**Pass**	0.4
`csv-parser.test.ts`	the same row always gives the same 32-character hash	**Pass**	0.3
`csv-parser.test.ts`	a file without a recognisable header is refused	**Pass**	0.5
`emi.test.ts`	EMI for ₹50,00,000 at 8.5% over 240 months is ₹43,391.16	**Pass**	1.0
`emi.test.ts`	schedule principal parts add up exactly to the principal and end at 0.00 (NFR-02)	**Pass**	1.3
`emi.test.ts`	odd amounts and short tenures still close at exactly 0.00	**Pass**	0.3
`emi.test.ts`	due dates clamp to the end of short months	**Pass**	0.2
`file-store.test.ts`	detects allowed types from the first bytes	**Pass**	1.2
`file-store.test.ts`	rejects GIF, executables and text even if renamed	**Pass**	0.2
`file-store.test.ts`	saves under a random name and refuses path tricks	**Pass**	7.9
`format.test.ts`	formatINR uses Indian digit grouping	**Pass**	15.4
`format.test.ts`	formatINR keeps two decimals for paise	**Pass**	0.3
`fy.test.ts`	31 March belongs to the earlier FY, 1 April to the later one (AC-22)	**Pass**	1.0
`fy.test.ts`	labels and ranges	**Pass**	1.0
`fy.test.ts`	tax hints are capped at the section limits	**Pass**	0.3
`fy.test.ts`	no qualifying loans → no hints	**Pass**	0.1
`news.test.ts`	parses items, decodes CDATA/entities/HTML and skips untitled items	**Pass**	5.2
`news.test.ts`	detects rate cuts, hikes, "from … to …" and unchanged	**Pass**	0.8
`news.test.ts`	categories	**Pass**	0.3
`news.test.ts`	a +25 bps change raises the EMI on a floating loan (AC-25)	**Pass**	0.5
`news.test.ts`	decode handles numeric entities	**Pass**	0.2
`pdf-statement.test.ts`	PAN keeps only the last four characters	**Pass**	1.1
`pdf-statement.test.ts`	amounts use Rs. and Indian grouping	**Pass**	15.6
`pdf-statement.test.ts`	builds a PDF that shows the masked PAN and never the full PAN (SR-09)	**Pass**	29.5
`prepayment.test.ts`	reduce tenure keeps the EMI and finishes earlier	**Pass**	1.5
`prepayment.test.ts`	reduce EMI keeps the end date and lowers the EMI	**Pass**	0.4

File	Test	Result	ms
prepayment.test.ts	reducing tenure saves more interest than reducing EMI	Pass	0.9
prepayment.test.ts	principal in the new schedule adds up to the new outstanding	Pass	0.3
prepayment.test.ts	amount must be positive and less than the outstanding	Pass	0.6
projection.test.ts	no loans and no growth: net worth rises by the surplus each month	Pass	1.2
projection.test.ts	growth compounds monthly	Pass	0.2
projection.test.ts	liabilities follow the EMI schedule and reach zero	Pass	0.2
projection.test.ts	at 0% growth, closing a loan now gains exactly its remaining interest	Pass	0.4
receipt-fields.test.ts	pharmacy receipt: merchant, grand total (not sub total) and date	Pass	3.2
receipt-fields.test.ts	skips "TAX INVOICE" header and reads Net Amount with ₹ and Indian commas	Pass	0.8
receipt-fields.test.ts	no total label: falls back to the largest amount	Pass	0.3
receipt-fields.test.ts	unreadable text gives nulls instead of guesses	Pass	0.3
recommend.test.ts	rate moves from max_rate at min_score to min_rate at 900	Pass	1.1
recommend.test.ts	ineligible products carry a reason and no rank (AC-20)	Pass	0.6
recommend.test.ts	eligible products are ranked by total cost, cheapest first	Pass	1.1
recommend.test.ts	amount and tenure limits are checked	Pass	15.5
security.test.ts	SR-01: passwords are bcrypt-hashed with cost 12, never stored as plain text	Pass	1491.0
security.test.ts	SR-02: a tampered or wrongly signed session token is rejected	Pass	5.0
security.test.ts	SR-02: a short JWT secret is refused	Pass	1.3
security.test.ts	SR-06: ids from the URL must be positive integers	Pass	0.8
security.test.ts	SR-08: unexpected errors return a generic 500 without internal details	Pass	6.3
security.test.ts	SR-08: database constraint errors map to safe messages	Pass	0.5
security.test.ts	SR-06: values rejected by the database (impossible date, out-of-range number) map to 422	Pass	1.3
security.test.ts	SR-09: PAN is masked on statements	Pass	0.2
upload.test.ts	accepts an allowed type and reports its detected type	Pass	3.4
upload.test.ts	missing or empty file → 422	Pass	0.9
upload.test.ts	too large → 413 before the content is inspected	Pass	0.2
upload.test.ts	content not in the allow-list → 415, even with an allowed-looking name	Pass	0.4
validation.test.ts	sign-up: password length 7/8, PAN format, email normalised (TC-003, TC-004)	Pass	4.4
validation.test.ts	asset: valuation not before purchase; values ≥ 0; known types only (TC-011, TC-012)	Pass	1.3
validation.test.ts	loan: rate 0 < r ≤ 60 and tenure 1–480 (TC-016)	Pass	0.7
validation.test.ts	transaction: amount > 0, ISO date, known type; filter limit ≤ 500 (TC-027)	Pass	0.7
validation.test.ts	import commit: 1–5000 rows, income/expense only (FR-14)	Pass	77.3
validation.test.ts	rules, receipts, vault, prepayment, projection and recommendation schemas	Pass	3.5

3. Database integration test suite

Command: `npx tsx --test --test-concurrency=1 "tests/db/**/*.test.ts"`

Test file	Covers	Tests	Passed	Failed	Time (ms)
advisor.test.ts	FR-26 advisor	2	2	0	62
atomicity.test.ts	NFR-03 fault injection	3	3	0	72

Test file	Covers	Tests	Passed	Failed	Time (ms)
credit-score.test.ts	FR-19 credit score	2	2	0	123
imports.test.ts	FR-14 import	4	4	0	73
loans-edit.test.ts **(new)**	FR-08 / FR-22 loans and FY list	4	4	0	117
news-refresh.test.ts **(new)**	FR-25 / NFR-04 feed refresh	2	2	0	56
news.test.ts	FR-25 news parsing	3	3	0	53
notifications.test.ts	FR-24 reminders	2	2	0	70
performance.test.ts	NFR-01 response time	1	1	0	400
prepayment.test.ts	FR-11 prepayment	4	4	0	197
projection.test.ts	FR-21 projection	2	2	0	85
receipts.test.ts	FR-16 receipts / OCR	2	2	0	978
recommend.test.ts	FR-20 recommendations	2	2	0	56
reports.test.ts	FR-22 FY report	2	2	0	101
rules.test.ts	FR-15 rules	4	4	0	53
schema.test.ts	Schema constraints (SR-06)	3	3	0	55
search.test.ts	FR-13 search	2	2	0	40
statement.test.ts	FR-23 statement	1	1	0	145
transaction-links.test.ts	FR-05 / FR-08 ledger links	2	2	0	48
users-assets.test.ts **(new)**	FR-01 / FR-05 / FR-06 / FR-12	4	4	0	78
vault.test.ts	FR-17 vault / NFR-04	3	3	0	63

3. Database integration test suite – individual results

File	Test	Result	ms
advisor.test.ts	advisor sees only assigned clients; other users are 403 (AC-26)	**Pass**	57.1
advisor.test.ts	only advisor accounts can be chosen	**Pass**	4.7
atomicity.test.ts	loan creation: a schedule failure leaves no loan behind	**Pass**	35.8
atomicity.test.ts	EMI payment: a ledger failure leaves the installment unpaid	**Pass**	10.3
atomicity.test.ts	prepayment: a failure at the last step leaves schedule, ledger and asset unchanged	**Pass**	25.9
credit-score.test.ts	score is 300–900 with 5 factors and the disclaimer (FR-19)	**Pass**	43.5
credit-score.test.ts	an overdue installment lowers the score; reading twice adds no history row (AC-19)	**Pass**	79.8
imports.test.ts	preview suggests categories and saves nothing (FR-14)	**Pass**	38.0
imports.test.ts	importing the same statement twice adds nothing the second time	**Pass**	17.3
imports.test.ts	a failing row rolls back the whole batch (NFR-03)	**Pass**	5.9
imports.test.ts	one user's import does not mark another user's rows as duplicates	**Pass**	12.1
loans-edit.test.ts	editing terms regenerates the schedule while nothing is paid (FR-08)	**Pass**	69.8
loans-edit.test.ts	editing is refused after a payment or a prepayment (409); other users get null (FR-08, FR-11)	**Pass**	27.5
loans-edit.test.ts	status change and delete are owner-only (FR-08, SR-05)	**Pass**	10.8
loans-edit.test.ts	FY list includes years with paid EMIs and the current FY (FR-22)	**Pass**	9.2
news-refresh.test.ts	refresh stores parsed articles and is idempotent	**Pass**	43.7
news-refresh.test.ts	feed errors become 503 and the news page falls back without calling the feed again	**Pass**	12.8
news.test.ts	+25 bps article: EMI change for floating loans only (AC-25)	**Pass**	44.6
news.test.ts	cached articles are listed and can be filtered by category	**Pass**	5.3
news.test.ts	unknown article is 404	**Pass**	3.1
notifications.test.ts	reminders are created once per installment and kind (AC-24)	**Pass**	50.0
notifications.test.ts	mark one read lowers unread by one; mark all sets it to 0	**Pass**	20.5

File	Test	Result	ms
performance.test.ts	dashboard and ledger search p90 under 2 seconds with 3,000 transactions (NFR-01)	Pass	399.7
prepayment.test.ts	simulate saves nothing and shows both options (FR-11)	Pass	63.4
prepayment.test.ts	apply reduce-tenure: paid rows untouched, outstanding drops by the amount, ledger entry written	Pass	57.3
prepayment.test.ts	apply funded from an asset deducts the asset; too-small asset is refused and nothing changes	Pass	46.3
prepayment.test.ts	amount must be below outstanding; other user gets 404	Pass	29.8
projection.test.ts	projection starts from today's assets and loans (FR-21)	Pass	65.8
projection.test.ts	closing another user's loan is a 404	Pass	19.4
receipts.test.ts	OCR reads the sample receipt (FR-16, NFR-01: under 10 s)	Pass	941.8
receipts.test.ts	confirm saves an expense, files the image, and refuses the same receipt again	Pass	35.7
recommend.test.ts	home loan comparison returns the catalogue ranked, ineligible ones with reasons (FR-20)	Pass	47.3
recommend.test.ts	a too-long tenure makes vehicle products ineligible	Pass	8.3
reports.test.ts	31 March and 1 April land in different financial years (AC-22)	Pass	50.9
reports.test.ts	home loan EMIs paid in the FY produce 80C and 24(b) hints	Pass	50.0
rules.test.ts	system rules categorise common merchants; unknown stays Uncategorized	Pass	25.9
rules.test.ts	a user's own rule beats the system rule only for that user (FR-15)	Pass	5.2
rules.test.ts	adding a rule re-categorises existing matching transactions	Pass	12.5
rules.test.ts	system rules cannot be deleted by a user	Pass	9.1
schema.test.ts	all core tables exist after migrations	Pass	28.7
schema.test.ts	emails are unique and stored in lower case	Pass	7.3
schema.test.ts	money cannot be negative (CHECK constraint)	Pass	19.4
search.test.ts	full-text search matches word stems and quoted phrases (FR-13)	Pass	33.5
search.test.ts	search never returns another user's transactions (SR-05)	Pass	6.1
statement.test.ts	statement has summary, assets, active loans and FY to date; PDF is produced (FR-23)	Pass	145.5
transaction-links.test.ts	deleting an asset or loan keeps its transactions with the link cleared (FR-05, FR-08)	Pass	44.0
transaction-links.test.ts	a transaction cannot point at an asset that does not exist	Pass	3.8
users-assets.test.ts	users: create, find by email, unique email (FR-01)	Pass	43.8
users-assets.test.ts	health check reads the database time	Pass	8.3
users-assets.test.ts	assets: revaluation adds history, rename does not; other users get nothing (FR-05, FR-06, SR-05)	Pass	19.0
users-assets.test.ts	transactions: only the owner can delete (FR-12, SR-05)	Pass	7.0
vault.test.ts	upload, find by tag and text, then delete (FR-17)	Pass	51.0
vault.test.ts	another user can't see, download, delete or link to my things	Pass	8.9
vault.test.ts	vault reports unavailable (503) when storage cannot be written (NFR-04)	Pass	2.7

4. Defects

No test failed in this run. Defects DEF-01 – DEF-03 found by the system validation run of 31-10-2026 were fixed in PR #77 and each fix has a regression test (`rules` API via system validation TC-033, `transaction-links.test.ts`, `security.test.ts` SR-06 case). See the System Validation Report.

Sign-off

Role	Name	Signature / Date
Test Lead	Likith Kumar	
Security & Traceability Lead	Vasu Tripathi	
Architecture Lead	Nishant Nag	
Requirements / UAT Lead	Madhusudhan P	